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Physics-Informed Neural Power Flow Solver for Three-Phase Unbalanced Distribution Networks

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***Abstract*—Power flow analysis is fundamental for the reliable operation and planning of modern distribution networks (DNs). However, conventional iterative numerical solvers are often computationally intensive, especially for large networks. In this work, we propose a Physics-Informed Graph Neural Network (PIGNN-3Φ) approach to speed-up the power flow calculations in three-phase unbalanced DNs. The proposed method directly exploits the network topology and learns the mapping from power injection to bus voltages while respecting the underlying physical laws. Physical consistency is enforced by embedding Kirchhoff's laws and Ohm's law relations into the model objective function, promoting only feasible and interpretable solutions while reducing dependence on large, high-quality labeled datasets typical of black-box deep learning approaches. Furthermore, we quantify model predictive uncertainty using the Monte Carlo dropout technique, which approximates Bayesian inference in neural networks and provides well-calibrated estimates of the model's epistemic uncertainty. The results show that the proposed PIGNN-3Φ outperforms black-box models and achieves an approximately $50\times$ computational gain over traditional numerical PF solvers while achieving comparable accuracy. Furthermore, the framework enables risk-aware decision making by providing calibrated predictive intervals for voltage estimates, enabling system operators to assess probabilities of constraint violations.**

***Index Terms*—Power flow analysis, physics-informed machine learning, distribution networks, uncertainty quantification.**

I. Introduction

The primary objective of power flow (PF) analysis in distribution networks (DNs) is to determine the bus voltages and line flows under specified loading conditions. Distribution system operators rely on these calculations to identify potential bottlenecks within their network and make informed decisions regarding grid planning and network upgrades. However, conventional numerical PF solvers, such as the Newton–Raphson method [1], are computationally intensive, particularly for performing large-scale planning studies, such as hosting capacity analysis, which often require millions of iterative PF calculations. Furthermore, even small modifications in grid topology necessitate a full recalculation of the numerical PF solution, further compounding the computational challenges.

Deep learning (DL)-based approaches offer a promising alternative to accelerate PF calculations by learning non-linear patterns from historical data [2], [3]. However, their black-box nature and reliance on extensive training datasets often limit their ability to model the intricate dynamics of electrical networks. Graph Neural Networks (GNNs) have emerged as a powerful tool for PF analysis, leveraging the inherent graph like structure of power grids to model diverse topologies [4]. Through message-passing, GNNs enable buses to exchange information with their neighbors, capturing complex electrical interactions within the network. For instance, the study in [5] employs a Graph Attention Network (GAT) to enhance graph-based learning by assigning weights to nodes through an attention mechanism, which effectively prioritizes critical nodes. However, most of the existing DL approaches are physics-agnostic, and their accuracy heavily depends on the quantity and quality of the training data. This underscores the need for hybrid approaches that integrate domain knowledge into the neural networks to ensure robustness and accuracy.

Physics-Informed Neural Networks (PINNs) are gaining increased attention in the scientific research community as hybrid approaches that combine data-based learning with underlying physical principles. Recent works, such as in [6], highlight the potential of PINNs to accelerate PF computations with improved generalization performance. Additionally, PINNs have also been utilized for other power system-related tasks such as line parameter estimation, state estimation, and fault detection [7], [8]. The study in [9] incorporates Kirchhoff's laws and system topology as auxiliary constraints to enhance PF accuracy and robustness. Similarly, the authors in [10] proposed a specialized GNN architecture, termed PowerFlowNet, which demonstrated comparable performance to the traditional Newton-Raphson method while significantly reducing the computational speed. However, most of the existing PINN-based methods for PF analysis are tailored for transmission networks, which differ significantly from DNs characterized by their radial topology, high R/X ratios, and unbalanced phase loading. Some approaches oversimplify the problem by focusing only on single-phase or balanced three-phase networks, thereby neglecting the complexities of modeling phase unbalance inherent in real-world networks. Therefore, there is a pressing need for better physics-aware methodologies that can accurately perform PF calculations in unbalanced DNs while maintaining computational efficiency.

Uncertainty quantification (UQ) is vital for PF analysis as it quantifies confidence in model predictions and enables risk-aware decision-making under the high penetration of distributed energy resources and fluctuating demand. Monte

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基于物理信息的神经网络三相不平衡配电网络潮流求解器

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摘要——潮流分析是现代配电网可靠运行和规划的基础。然而，传统的迭代数值求解器计算量通常很大，尤其是在处理大型网络时。本文提出了一种基于物理信息的图神经网络（PIGNN-3Φ）方法来加速三相不平衡配电网的潮流计算。该方法直接利用网络拓扑结构，学习从功率注入到母线电压的映射关系，同时遵循物理定律。通过将基尔霍夫定律和欧姆定律嵌入模型目标函数，确保了物理一致性，从而只生成可行且可解释的解，同时减少了对大型高质量标注数据集的依赖，而这正是黑盒深度学习方法的典型特征。此外，我们使用蒙特卡罗 dropout 技术量化模型的预测不确定性，该技术近似于神经网络中的贝叶斯推理，并能提供模型认知不确定性的良好校准估计。结果表明，所提出的 PIGNN-3Φ模型优于黑箱模型，在保持相当精度的同时，计算量比传统数值 PF 求解器提高了约 50 倍。此外，该框架通过提供校准后的电压估计预测区间，支持风险感知决策，使系统运行人员能够评估约束违反的概率。

索引词——潮流分析、基于物理的机器学习、配电网络、不确定性量化。

一、配电网潮流（PF）分析的主要目标是在特定负荷条件下确定母线电压和线路潮流。配电系统运营商依靠这些计算来识别网络中的潜在瓶颈，并就电网规划和网络升级做出明智的决策。然而，传统的数值潮流求解器，例如牛顿-拉夫逊法[1]，计算量巨大，尤其是在进行大规模规划研究（例如容量分析）时，通常需要数百万次的迭代潮流计算。此外，即使电网拓扑结构发生微小变化，也需要重新计算数值潮流解，这进一步加剧了计算的挑战。

基于深度学习（DL）的方法通过学习历史数据中的非线性模式，为加速 PF 计算提供了一种很有前景的替代方案[2][3]。然而，它们的黑箱特性以及对大量训练数据集的依赖性通常这限制了它们对复杂电网动态特性的建模能力。图神经网络（GNN）已成为功率因数分析的强大工具，它利用电网固有的图状结构来模拟各种拓扑结构[4]。通过消息传递，GNN 使母线能够与其邻居交换信息，从而捕捉网络内部复杂的电气交互。例如，文献[5]采用图注意力网络（GAT）通过注意力机制为节点分配权重来增强基于图的学习，从而有效地优先处理关键节点。然而，大多数现有的深度学习方法与物理特性无关，其准确性严重依赖于训练数据的数量和质量。这凸显了将领域知识集成到神经网络中以确保鲁棒性和准确性的混合方法的必要性。

物理信息神经网络（PINN）作为一种将数据驱动学习与底层物理原理相结合的混合方法，在科学研究领域日益受到关注。近期研究，例如文献[6]，强调了 PINN 在加速功率因数（PF）计算和提升泛化性能方面的潜力。此外，PINN 还被用于其他电力系统相关任务，例如线路参数估计、状态估计和故障检测[7][8]。文献[9]将基尔霍夫定律和系统拓扑结构作为辅助约束，以提高功率因数的精度和鲁棒性。类似地，文献[10]提出了一种名为 PowerFlowNet 的专用图神经网络（GNN）架构，该架构在显著降低计算速度的同时，展现出与传统牛顿-拉夫逊法相当的性能。然而，大多数现有的基于 PINN 的功率因数分析方法都是针对输电网络设计的，而输电网络与配电网（DN）存在显著差异，配电网的特点是辐射状拓扑结构、高 R/X 比和不平衡相负载。一些方法过于简化问题，仅关注单相或平衡的三相网络，从而忽略了实际网络中固有的相间不平衡建模的复杂性。因此，迫切需要更完善的、能够考虑物理因素的方法，以便在保持计算效率的同时，准确地对不平衡配电网进行功率因数计算。

不确定性量化（UQ）对于电力函数分析至关重要，因为它量化了模型预测的置信度，并能够在分布式能源高渗透率和需求波动的情况下实现风险感知决策。

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Carlo (MC) dropout [11] provides a scalable approach for obtaining probabilistic outputs from neural networks, offering a practical alternative to computationally intensive Bayesian inference. The study in [12] employs MC dropout within a GNN model to enable probabilistic voltage predictions with quantified uncertainty. Despite their potential, the application of probabilistic PINNs in power systems studies remains relatively underexplored and requires further investigation.

Building upon our prior work in [6], this paper extends the application of PINNs for solving the probabilistic PF problem in three-phase unbalanced DNs. We present a novel physics-informed graph neural network (PIGNN-3Φ) framework that combines the computational efficiency of data-driven methods with the physical principles based on PF equations. This ensures accurate and physically consistent PF solutions. The key contributions of this work are:

- A probabilistic PIGNN-3Φ model for performing the accurate PF calculations in three-phase unbalanced DNs.
- A physics-informed loss function incorporating Kirchhoff's law (KCL) and Ohm's law as soft constraints to ensure physical consistency and feasible solutions.
- MC dropout-enabled probabilistic inference for quantifying uncertainty in model predictions.

The remainder of this paper is organized as follows: Section II provides an overview of the PF formulation for radial DNs. In Section III, we discuss the proposed methodology in detail. Experimental results are provided in Section IV, followed by the conclusion in Section V.

II. Power Flow Formulation

The DistFlow model, as described in [13], provides an accurate representation of power flows in radial DNs. In this section, we discuss the DistFlow formulation and its linearized form for three-phase unbalanced systems (i.e., LinDist3Flow). Let $\mathcal{N} = \{0, 1, \ldots, n\}$ denote the set of buses, and $\mathcal{E} \subseteq \mathcal{N} \times \mathcal{N}$ be the set of electrical lines in the simplified radial DN, as shown in Fig. 1. For each bus $n \in \mathcal{N}$, the squared voltage is denoted as $v_n = |V_n|^2$. For each line $(\pi_n, n) \in \mathcal{E}$, the line impedance is $z_n = r_n + jx_n$, with resistance r_n and reactance x_n. The complex power injection at bus n is $s_n = p_n + jq_n$, and the squared current magnitude is $\ell_n = |I_n|^2$.

The non-linear DistFlow equations are expressed as:

$$v_n = |V_n|^2, \qquad \ell_n = |I_n|^2 \tag{1}$$

$$\sum\nolimits_{k:n\to k} P_k = p_n + P_n - r_n \ell_n \tag{2}$$

$$\sum\nolimits_{k:n\to k} Q_k = q_n + Q_n - x_n \ell_n \tag{3}$$

$$v_n = v_{\pi_n} - 2(r_n P_n + x_n Q_n) + (r_n^2 + x_n^2)\ell_n \tag{4}$$

$$v_{\pi_n}\, \ell_n = P_n^2 + Q_n^2 \tag{5}$$

The equations (2)-(5) represent the nodal power balance and voltage drop relations in a single-phase radial network. This formulation is non-linear due to the quadratic relation in (5). By neglecting the loss terms $r_n\ell_n$, $x_n\ell_n$, and $(r_n^2 + x_n^2)\ell_n$,

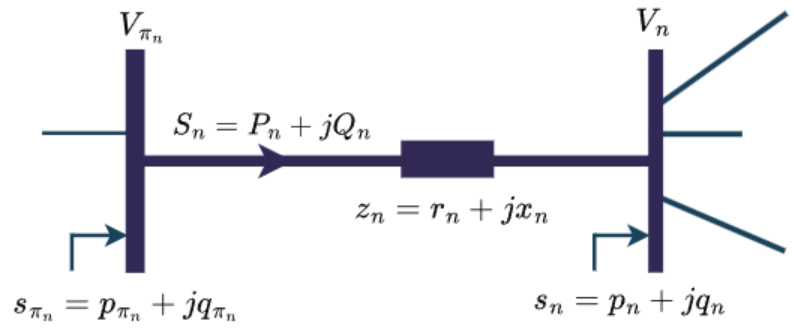


Fig. 1: A simple two-bus distribution system where an electrical line with impedance z_n connects bus n to bus π_n.

the model reduces to its linearized form (i.e., LinDistFlow). This linearized formulation is widely used in planning and control studies as it offers improved analytical tractability while introducing only minor approximation errors. For three-phase unbalanced systems, these equations can be expressed in a compact matrix notation, where phase-wise voltage and line power flow variables are defined as:

$$\mathbf{V} \in \mathbb{R}^{3|\mathcal{N}|}, \quad \mathbf{P}, \mathbf{Q} \in \mathbb{R}^{3|\mathcal{E}|}.$$

For each branch $(i, j) \in \mathcal{E}$, we introduce the parameter $\kappa = \sqrt{3}$ and define the per-branch resistance and reactance coupling matrices as:

$$\mathbf{M}_R^{(ij)} = \begin{bmatrix} R_{aa} & R_{ab} - \kappa X_{ab} & R_{ac} + \kappa X_{ac} \\ R_{ba} + \kappa X_{ba} & R_{bb} & R_{bc} - \kappa X_{bc} \\ R_{ca} - \kappa X_{ca} & R_{cb} + \kappa X_{cb} & R_{cc} \end{bmatrix},$$

$$\mathbf{M}_X^{(ij)} = \begin{bmatrix} X_{aa} & X_{ab} + \kappa R_{ab} & X_{ac} - \kappa R_{ac} \\ X_{ba} - \kappa R_{ba} & X_{bb} & X_{bc} + \kappa R_{bc} \\ X_{ca} + \kappa R_{ca} & X_{cb} - \kappa R_{cb} & X_{cc} \end{bmatrix}.$$

The per-branch coupling matrices are then assembled into the following block-diagonal matrices:

$$\mathbf{M}_R := \mathrm{blkdiag}\{\mathbf{M}_R^{(ij)}\}, \quad \mathbf{M}_X := \mathrm{blkdiag}\{\mathbf{M}_X^{(ij)}\}.$$

Equivalently, the LinDist3Flow voltage drop (Ohm's law) and power balance equations can be written in compact matrix form using the branch–bus incidence matrix $\mathbf{A}$ as:

$$(\mathbf{A} \otimes \mathbf{I}_3)\mathbf{V} = 2\mathbf{M}_R\mathbf{P} + 2\mathbf{M}_X\mathbf{Q}, \tag{6}$$

$$(\mathbf{A}^\top \otimes \mathbf{I}_3)\mathbf{P} = \mathbf{p}_{\mathrm{net}}, \quad (\mathbf{A}^\top \otimes \mathbf{I}_3)\mathbf{Q} = \mathbf{q}_{\mathrm{net}}, \tag{7}$$

where $\mathbf{p}_{\mathrm{net}}, \mathbf{q}_{\mathrm{net}} \in \mathbb{R}^{3|\mathcal{N}|}$ denote the per-phase net injections. The Kronecker product $\mathbf{A} \otimes \mathbf{I}_3$ extends the branch–bus incidence matrix $\mathbf{A} \in \mathbb{R}^{|\mathcal{E}|\times|\mathcal{N}|}$ to three phases, effectively modeling mutual phase coupling within each line segment. This formulation provides a computationally efficient and convex representation that is analytically tractable for large radial networks and sufficiently accurate under moderate voltage drops and line losses. To ensure physical consistency in the proposed PIGNN-3Φ model's predictions, we embed the LinDist3Flow relations (6)–(7) into the training objective as soft constraints. This helps the model learn a physically plausible PF solution by penalizing violations of Ohm's law and the power balance equations. This physics-guided regularization improves model generalization to out-of-distribution operating conditions and increases its robustness to measurement noise during training.

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卡洛（MC）dropout [11] 为从神经网络中获取概率输出提供了一种可扩展的方法，为计算密集型的贝叶斯推断提供了一种实用的替代方案。文献 [12] 在 GNN 模型中应用了 MC dropout，实现了具有量化不确定性的概率电压预测。尽管概率 PINN 具有潜力，但其在电力系统研究中的应用仍相对较少，需要进一步研究。

本文基于我们在[6]中的前期工作，扩展了物理信息图神经网络（PINN）在三相不平衡配电网概率功率因数（PF）问题求解中的应用。我们提出了一种新型的物理信息图神经网络（PIGNN-3Φ）框架，该框架结合了数据驱动方法的计算效率和基于 PF 方程的物理原理，从而确保了 PF 解的准确性和物理一致性。本文的主要贡献如下：

- 用于在三相不平衡配电网络中进行精确功率因数计算的概率 PIGNN-3Φ 模型。
- 一种结合了基尔霍夫定律（KCL）和欧姆定律的物理信息损失函数，作为软约束，以确保物理一致性和可行解。
- MC dropout 支持的概率推断用于量化模型预测中的不确定性。

本文余下部分组织如下：第二节概述了径向配电网的 PF 公式；第三节详细讨论了所提出的方法；第四节给出了实验结果；第五节总结全文。

二、PFF

如文献[13]所述，DistFlow 模型能够精确表示辐射状配电网中的功率潮流。本节将讨论 DistFlow 模型及其在三相不平衡系统中的线性化形式（即 LinDist3Flow）。令 N = {0, 1, ..., n}表示母线集合，E ⊆ N × N 表示简化辐射状配电网中的电力线路集合，如图 1 所示。对于每个母线 n ∈ N，电压平方记为 v = |V|。对于每条线路(π, n) ∈ E，线路阻抗为 z = r + jx，其中 r 为电阻，x 为电抗。母线 n 处的复功率注入为 s = p + jq，电流幅值的平方记为|I|。

非线性 DistFlow 方程表示如下：

$$v = |V|, = |I| \tag{1}$$

$$\sum_{k:n\to k} P = p + P - r \tag{2}$$

$$\sum_{k:n\to k} Q = q + Q - x \tag{3}$$

$$v = v - 2(rP + xQ) + (r + x) \tag{4}$$

$$v = P + Q \tag{5}$$

方程 (2)-(5) 表示单相辐射状网络中的节点功率平衡和电压降关系。由于 (5) 中的二次关系，该公式是非线性的。忽略损耗项 r、x 和 (r+x)，

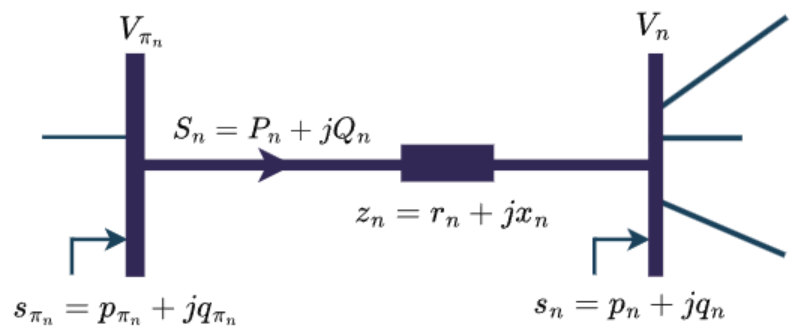


图 1：一个简单的双母线配电系统，其中阻抗为 z 的电力线连接母线 n 和母线 π。

该模型可简化为线性化形式（即 LinDistFlow）。这种线性化形式广泛应用于规划和控制研究中，因为它在引入近似误差的同时，提供了更优异的解析可解性。对于三相不平衡系统，这些方程可以用紧凑的矩阵形式表示，其中各相电压和线路功率流变量定义如下：

$$V \in R, \qquad P, Q \in R。$$

对于每个分支 (i, j) ∈ E，我们引入参数 κ = 1/√3. 将各支路的电阻和电抗耦合矩阵定义为：

$$先生 = \begin{bmatrix} RR - \kappa X & R + \kappa X & \\ R + \kappa X & RR - \kappa X & \\ R - \kappa X & R + \kappa X & R \end{bmatrix},$$

$$MX = \begin{bmatrix} X & X + \kappa R & X - \kappa R \\ X - \kappa R & X & X + \kappa R \\ X + \kappa R & X - \kappa R & X \end{bmatrix}$$

然后将各分支耦合矩阵组装成以下分块对角矩阵：

$$M := \text{blkdiag}\{MR\}, \qquad M := \text{blkdiag}\{MX\}。$$

等价地，LinDist3Flow 电压降（欧姆定律）和功率平衡方程可以使用分支-母线关联矩阵 A 写成紧凑的矩阵形式：

$$(A \otimes I)V = 2MP + 2MQ, \tag{6}$$

$$(A \otimes I)P = p, \qquad (A \otimes I)Q = q, \tag{7}$$

其中 p, q∈R 表示每相净注入量。克罗内克积 A ⊗ I 将支路-母线关联矩阵 A ∈ R 扩展到三相，有效地模拟了每段线路内的相间耦合。该公式提供了一种计算高效且凸的表示，对于大型辐射状网络而言，其解析解易于处理，并且在中等电压降和线路损耗下具有足够的精度。为了确保所提出的 PIGNN-3Φ 模型预测的物理一致性，我们将 LinDist3Flow 关系式 (6)–(7) 作为软约束嵌入到训练目标中。这有助于模型通过惩罚违反欧姆定律和功率平衡方程的情况来学习一个物理上合理的功率因数解。这种物理引导的正则化提高了模型对配电网络外运行条件的泛化能力，并增强了其在训练过程中对测量噪声的鲁棒性。

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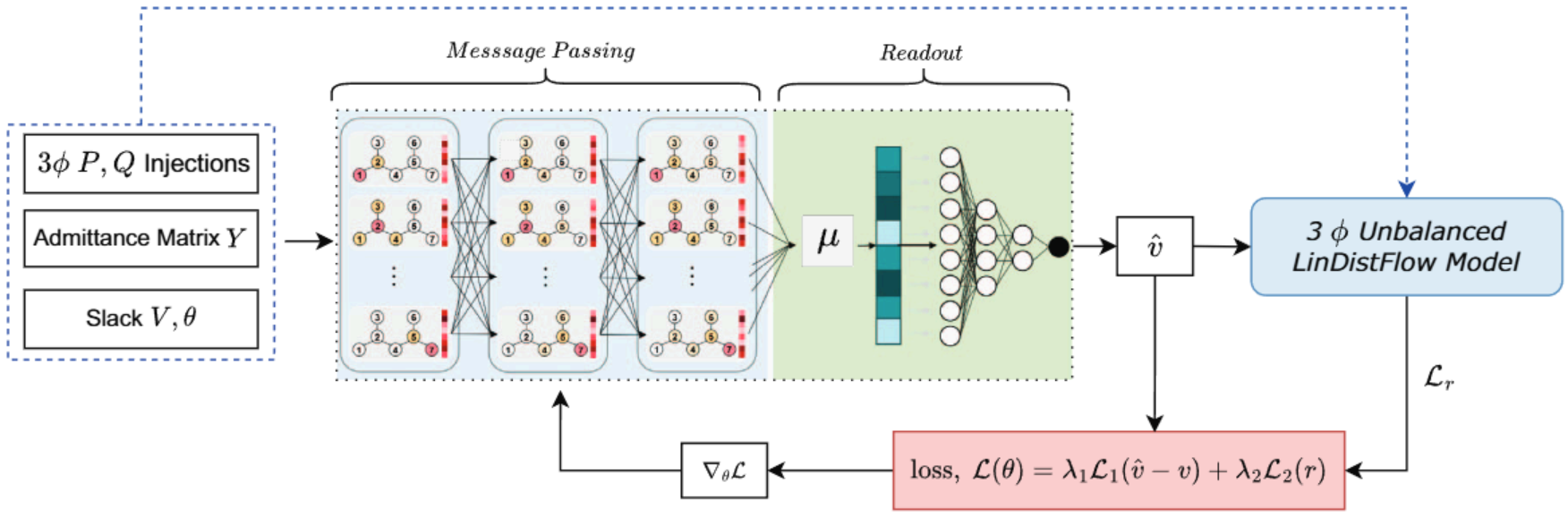


Fig. 2: Proposed PIGNN-3Φ model architecture. It processes three-phase input power injections and admittance matrix through GAT layers, leveraging message passing to learn complex node interactions. The readout stage outputs predicted voltages $\hat{v}$. To enforce the physical consistency, residuals $\mathcal{L}_r$ are added to the overall loss $\mathcal{L}(\theta)$, combining data-fit and physics constraints.

III. Proposed Methodology

In this study, we propose the PIGNN-3Φ model to predict probabilistic voltages in three-phase unbalanced DNs. The primary aim is to produce fast, accurate approximations of the three-phase unbalanced PF problem while providing UQ. By explicitly modeling phase unbalance and embedding the physical constraints into the learning objective, the framework achieves improved generalization to unseen operational conditions and substantially reduces computational cost compared to iterative solvers. Fig. 2 shows the overall architecture of the proposed PIGNN-3Φ model, which comprises two key components: (i) a graph-based architecture, and (ii) a physics-informed loss function that employs the LinDist3Flow equations. The proposed model takes per-phase power injections, the admittance matrix, and slack bus conditions as inputs to predict voltages at all non-slack buses. The architecture incorporates a multi layer GAT [5] mechanism to effectively model the influence among neighboring nodes during message passing, thereby capturing fine-grained phase-level interactions. The node embeddings, which represent the latent feature vectors, are aggregated via a readout module composed of Multi-Layer Perceptrons (MLPs)[1]. The readout maps the aggregated embeddings to the predicted per-phase voltage magnitudes $\hat{v}$. The model predictions are constrained by a physics-informed loss function, which seamlessly integrates physics-based residuals with the data-fit term to produce feasible solutions that adhere to the underlying physical principles.

A. Graph Attention Network (GAT)

We model the electrical distribution grid as an undirected graph $\mathcal{G} = (\mathcal{N}, \mathcal{E})$, where $\mathcal{N}$ represents the set of nodes (i.e., physical buses) and $\mathcal{E}$ denotes the set of edges (i.e., power lines). Each node $i \in \mathcal{N}$ is associated with a feature vector $\mathbf{x}_i \in \mathbb{R}^d$, encoding electrical quantities such as target voltage magnitude $|V_i|$ and input power injections (p_i, q_i). Each edge $(i, j) \in \mathcal{E}$ carries a feature vector $\mathbf{e}_{ij} \in \mathbb{R}^k$, typically including line parameters such as resistance R_{ij} and reactance X_{ij}. The underlying mechanism of graph-based learning is the *message-passing* process, in which nodes iteratively exchange information with their neighbors to update their embeddings via three steps: (i) message computation, (ii) aggregation, and (iii) node update. GAT employs attention coefficients to dynamically assign importance weights to neighboring nodes during the message aggregation process. This mechanism is particularly advantageous in electrical networks, where the influence of one bus on another can vary significantly depending on the network topology and loading conditions. In each GAT layer l, messages $\mathbf{m}_{ij}^{(l)}$ are passed from node j to node i to captures both node-level and edge-level dependencies. The raw (unnormalized) attention scores $e_{ij}^{(l)}$ are computed by a shared attention mechanism and normalized across node i's neighborhood using a softmax function to obtain attention weights $\alpha_{ij}^{(l)}$. These weights rescale the messages based on their importance, which are then aggregated into $\mathbf{s}_i^{(l)}$. Finally, a learnable update function $U^{(l)}(\cdot)$, typically an MLP, produces the new embedding $\mathbf{h}_i^{(l+1)}$. The mathematical formulations of these message passing steps for a layer l are defined as:

$$\mathbf{m}_{ij}^{(l)} = \psi^{(l)}\left(\mathbf{h}_j^{(l)}, e_{ij}\right), \tag{8}$$

$$e_{ij}^{(l)} = \mathrm{LeakyReLU}\left(\mathbf{a}^{(l)\top}\left[\,\mathbf{W}^{(l)}\mathbf{h}_i^{(l)} \,\|\, \mathbf{W}^{(l)}\mathbf{h}_j^{(l)}\,\right]\right), \tag{9}$$

$$\alpha_{ij}^{(l)} = \frac{\exp\left(e_{ij}^{(l)}\right)}{\sum_{k\in\mathcal{N}_i}\exp\left(e_{ik}^{(l)}\right)}, \tag{10}$$

$$\mathbf{s}_i^{(l)} = \sum_{j\in\mathcal{N}_i} \alpha_{ij}^{(l)}\,\mathbf{m}_{ij}^{(l)}, \tag{11}$$

$$\mathbf{h}_i^{(l+1)} = U^{(l)}\left(\mathbf{h}_i^{(l)}, \mathbf{s}_i^{(l)}\right), \tag{12}$$

where $\mathbf{W}^{(l)} \in \mathbb{R}^{d'\times d}$ is a learnable weight matrix mapping applied to node features and $\mathbf{a}^{(l)} \in \mathbb{R}^{2d'}$ is the trainable attention vector. To enhance the model's expressiveness, we employ a *multi-head attention* strategy. Instead of relying on

[1]An MLP is a feedforward neural network consisting of multiple layers.

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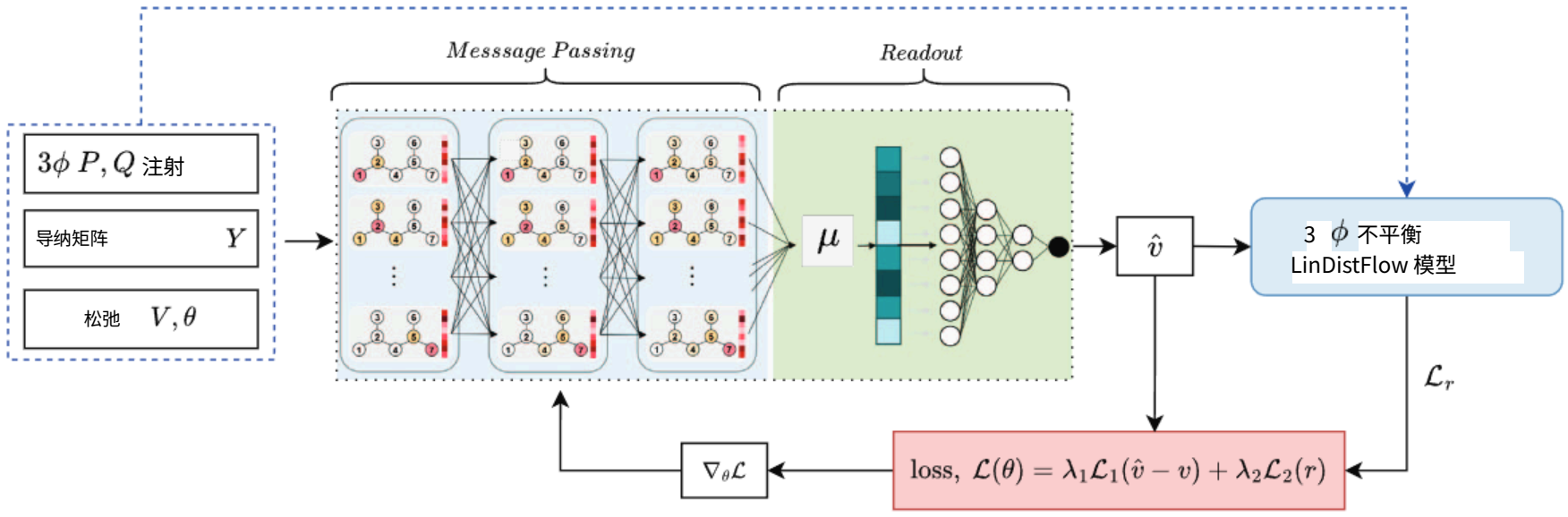


图 2：所提出的 PIGNN-3Φ 模型架构。它通过 GAT 层处理三相输入功率注入和导纳矩阵，并利用消息传递来学习复杂的节点交互。读出级输出预测电压 ˆv。为了确保物理一致性，残差 Lare 被添加到总损耗 L(θ) 中，从而结合数据拟合和物理约束。

三、下午

本研究提出了一种名为 PIGNN-3Φ的模型，用于预测三相不平衡配电网中的概率电压。其主要目标是在提供不确定性量化（UQ）的同时，快速、准确地近似求解三相不平衡功率因数问题。通过显式地对相不平衡进行建模，并将物理约束嵌入到学习目标中，该框架能够更好地泛化到未知的运行工况，并且与迭代求解器相比，显著降低了计算成本。图 2 展示了所提出的 PIGNN-3Φ模型的整体架构，该架构包含两个关键组件：（i）基于图的架构；（ii）采用 LinDist3Flow 方程的物理信息损失函数。该模型以每相功率注入、导纳矩阵和平衡母线条件作为输入，预测所有非平衡母线的电压。该架构集成了一个多层 GAT[5]机制，以有效地模拟消息传递过程中相邻节点之间的相互影响，从而捕捉细粒度的相级交互作用。代表潜在特征向量的节点嵌入通过由多层感知器（MLP）组成的读出模块进行聚合。该读出模块将聚合后的嵌入映射到预测的每相电压幅值ˆv。模型预测受到基于物理原理的损失函数的约束，该损失函数将基于物理原理的残差与数据拟合项无缝集成，从而生成符合底层物理原理的可行解。

A. 图注意力网络（GAT）

我们将配电网建模为一个无向图 G = (N, E)，其中 N 表示节点集（即物理母线），E 表示边集（即电力线）。每个节点 i ∈ N 都关联一个特征向量 x ∈ R，该向量编码了诸如目标电压幅值 |V| 和输入功率注入 (p, q) 等电量。每条边 节点 (i, j) ∈ E 携带特征向量 e∈ R，通常包含线路参数，例如电阻 R 和电抗 X。基于图的学习的底层机制是消息传递过程，其中节点通过三个步骤迭代地与其邻居交换信息来更新其嵌入：(i) 消息计算，(ii) 聚合，以及 (iii) 节点更新。GAT 使用注意力系数在消息聚合过程中动态地为邻居节点分配重要性权重。这种机制在电力网络中尤为有利，因为一个母线对另一个母线的影响会根据网络拓扑和负载条件而显著变化。在每个 GAT 层 l 中，消息从节点 j 传递到节点 i，以捕获节点级和边级依赖关系。原始（未归一化）注意力分数由共享注意力机制计算，并使用 softmax 函数在节点 i 的邻域内进行归一化，从而获得注意力权重 α。这些权重根据消息的重要性对其进行重新缩放，然后聚合为 s。最后，一个可学习的更新函数 U(·)（通常是一个多层感知器）生成新的嵌入 hi。第 l 层消息传递步骤的数学公式定义如下：

$$m = \psi\left(\text{他}\right), \tag{8}$$

$$e = \text{LeakyReLU}\left(a\left[Wh \,\|\, Wh\right]\right), \tag{9}$$

$$a = \sum_{k \in Nexp} \frac{\text{前任 e}}{\left(\overline{\quad}\right)}, \tag{10}$$

$$s = \sum_{j \in N} \text{嗯}, \tag{11}$$

$$\text{你好} = U\left(h, s\right), \tag{12}$$

其中 W∈Ris 是应用于节点特征的可学习权重矩阵映射，a∈Ris 是可训练的注意力向量。为了增强模型的表达能力，我们采用了一种多头注意力策略。我们并不依赖于……

[1] 多层感知器（MLP）是一种由多层组成的前馈神经网络。

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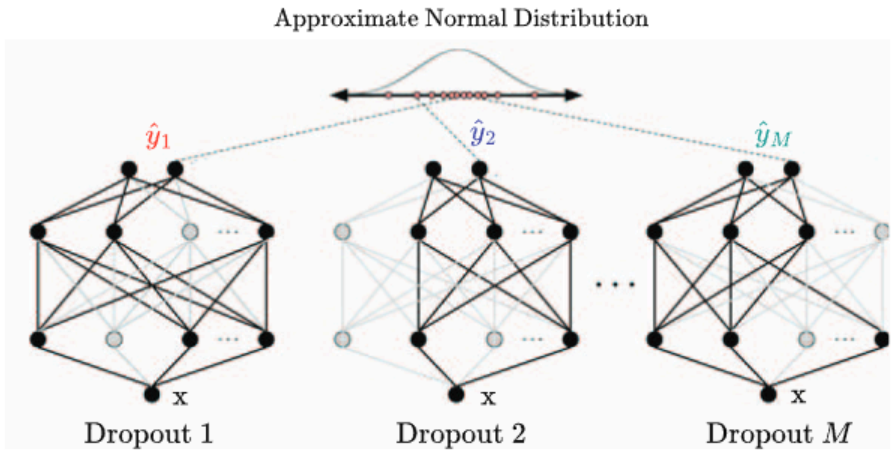


Fig. 3: Monte Carlo dropout process for uncertainty quantification. For a single input x, we take multiple stochastic predictions $\hat{y}$ to estimate the predictive mean and variance.

a single attention mechanism, multiple attention heads are applied in parallel, with each head capturing distinct and intricate relational patterns among neighboring nodes.

B. Physics-Informed Loss Function

A key component of the proposed model is its physics-informed loss function, which combines a data-fit term with a physics-based loss term derived from the LinDist3Flow equations. The constraint residuals are defined as:

$$\mathbf{r}_{\text{Ohm}} = \mathcal{A}\,\hat{\mathbf{v}} \;-\; 2\mathbf{M}_R\,\hat{\mathbf{P}} \;-\; 2\mathbf{M}_X\,\hat{\mathbf{Q}}, \tag{13}$$

$$\mathbf{r}^P_{\text{KCL}} = \mathcal{A}^\top\hat{\mathbf{P}} - \mathbf{p}_{\text{net}}, \qquad \mathbf{r}^Q_{\text{KCL}} = \mathcal{A}^\top\hat{\mathbf{Q}} - \mathbf{q}_{\text{net}}, \tag{14}$$

where $\mathcal{A} = \mathbf{A} \otimes \mathbf{I}_3$. The above constraint residuals are then combined to form a unified physics-based loss term:

$$\mathcal{L}_r = \frac{1}{3|\mathcal{E}|}\left\|\mathbf{r}_{\text{Ohm}}\right\|_2^2 + \frac{1}{3|\mathcal{N}|}\left(\left\|\mathbf{r}^P_{\text{KCL}}\right\|_2^2 + \left\|\mathbf{r}^Q_{\text{KCL}}\right\|_2^2\right), \tag{15}$$

The final objective function is the weighted sum of data-fit loss and physics residual terms:

$$\mathcal{L}(\theta) \;=\; \lambda_1\,\mathcal{L}_1(\hat{v} - v_{true}) \;+\; \lambda_2\,\mathcal{L}_{\text{r}}, \tag{16}$$

where $\lambda_1, \lambda_2 > 0$ are tunable hyperparameters that balance the contributions of the data-fit and the physics loss terms. This formulation ensures that the model remains faithful to the observed measurements while enforcing the three-phase PF constraints. Specifically, Ohm's law and the power balance constraints serve as a soft regularizer that restricts the model's feasible solution manifold and steers it toward physically admissible regimes. This enforces a principled trade-off between empirical fit and physical consistency, accelerating convergence and improving generalization. Moreover, the physics-based loss terms enable the network to capture cross-phase dependencies through the matrices $\mathbf{M}_R$ and $\mathbf{M}_X$.

C. Uncertainty Quantification Using MC Dropout

Bayesian inference provides a principled framework for modeling predictive uncertainty, which is essential for risk-aware operation in DNs. Quantifying uncertainty in predicted voltages enables operators to implement chance-constrained control and to assess the risk of voltage limit violations.

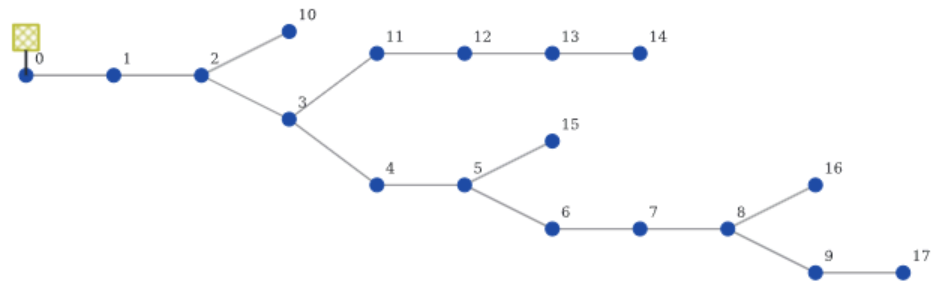


Fig. 4: Modified 18-bus CIGRE LV test network.

The MC dropout approach, as demonstrated in [11], approximates Bayesian inference in neural networks by interpreting dropout as a variational approximation to a deep Gaussian process. While dropout is traditionally used during model training to prevent overfitting by randomly deactivating a subset of hidden units, MC dropout extends this mechanism to inference. During inference, we approximate the posterior predictive distribution by performing M stochastic forward passes through the model, each employing an independently sampled dropout mask drawn from a Bernoulli distribution. This procedure yields an ensemble of voltage predictions for each input instance. The empirical predictive mean and variance are then computed to approximate the predictive distribution and to quantify model confidence. Fig. 3 illustrates this process, where M dropout masks are applied to output the predictive distribution of the target y for the same input x.

IV. Experimental Setup

We evaluated the proposed PIGNN-3Φ model on the 18-bus Modified CIGRE LV unbalanced distribution feeder, as detailed in [14]. The feeder layout is shown in Fig. 4. To generate the training dataset, we adopted an MC sampling strategy commonly used in PF studies. Specifically, active and reactive power profiles for residential customers were first sampled from a pool of standard load curves. To introduce realistic variability in the load patterns, the sampled profiles were perturbed with additive Gaussian noise (standard deviation 15%), applied independently across phases and time steps. Numerical PF simulations were subsequently performed using the `pandapower` framework [15] on the perturbed load profiles to obtain the corresponding voltage labels. We generated 2400 grid snapshots and split them into training, validation, and test sets using a $60/20/20$ ratio. The model was implemented in Python using the PyTorch and PyTorch Geometric libraries [16]. For benchmarking, we implemented two black-box neural network models: a GNN and a fully connected neural network (FCNN). Both baselines consist of 5 hidden layers with 128 units per layer and ReLU activations.

V. Numerical Results

The proposed PIGNN-3Φ model is compared against black-box GNN and FCNN models, using the Newton–Raphson PF solution as the baseline. The comparison evaluates four key aspects: (i) voltage prediction accuracy; (ii) calibration and quality of the MC dropout uncertainty estimates; (iii)

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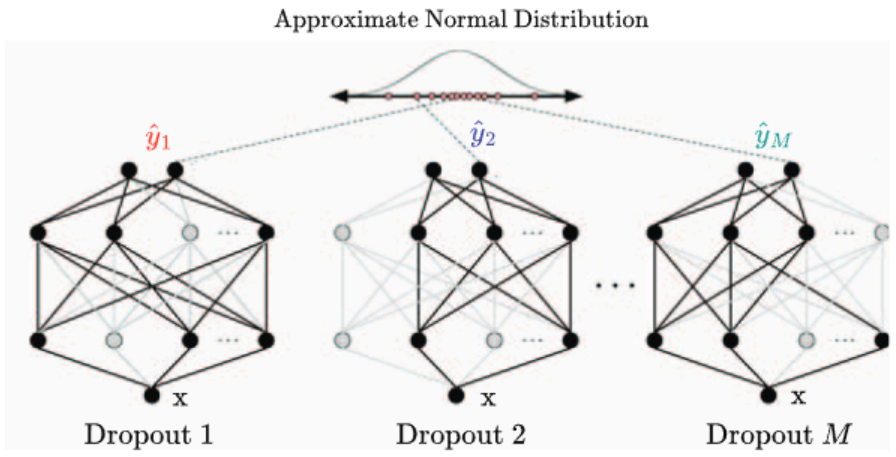


图 3：用于不确定性量化的蒙特卡罗 dropout 过程。对于单个输入 x，我们进行多次随机预测 ˆy 以估计预测均值和方差。单一的注意力机制，并行应用多个注意力头，每个注意力头捕获相邻节点之间独特而复杂的关联模式。

B. 基于物理学的损失函数

所提出的模型的一个关键组成部分是其物理信息损失函数，该函数将数据拟合项与源自 LinDist3Flow 方程的基于物理的损失项相结合。约束残差定义如下：

$$r = A\hat{v} - 2M \quad \hat{P} - 2M \quad 问, \tag{13}$$

$$r = A\hat{P} - p, \quad r = A\hat{Q} - q, \tag{14}$$

其中 A = A ⊗ I。上述约束残差组合起来，形成一个统一的基于物理的损失项：

$$L = \frac{1}{3|E|} \left\| r \right\| + \frac{1}{3|N|} \left(\left\| r \right\| + \left\| r \right\| \right), \tag{15}$$

最终目标函数是数据拟合损失项和物理残差项的加权和：

$$L(\theta) = \lambda L(\hat{v} - v) + \lambda L, \tag{16}$$

其中 λ，λ>0 是可调超参数，用于平衡数据拟合项和物理损失项的贡献。这种公式确保模型在满足三相功率函数约束的同时，保持对观测测量结果的忠实性。具体而言，欧姆定律和功率平衡约束作为软正则化项，限制了模型的可行解流形，并引导其趋向于物理上可接受的区域。这在经验拟合和物理一致性之间实现了合理的权衡，从而加速了收敛并提高了泛化能力。此外，基于物理的损失项使得网络能够通过矩阵 Mand M 捕捉跨相依赖性。

C. 使用 MC Dropout 进行不确定性量化

贝叶斯推断为预测不确定性建模提供了一个原则性的框架，这对于配电网的风险感知运行至关重要。量化预测电压的不确定性使运营商能够实施机会约束控制并评估电压限值超标的风险。

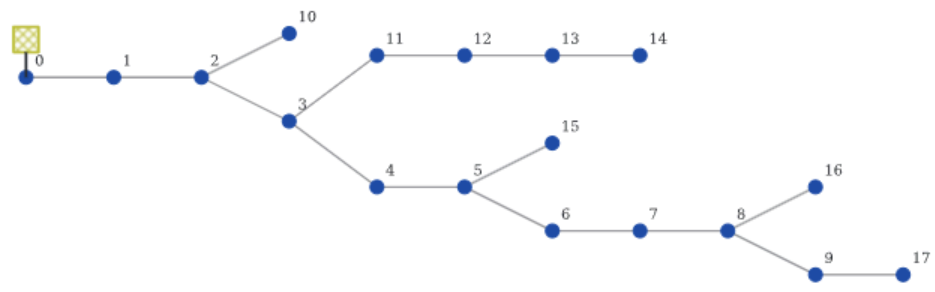


图 4：改进的 18 节点 CIGRE 低压测试网络。

如[11]所示，MC dropout 方法通过将 dropout 解释为深度高斯过程的变分近似，来近似神经网络中的贝叶斯推理。虽然 dropout 传统上用于模型训练期间，通过随机停用一部分隐藏单元来防止过拟合，但 MC dropout 将这种机制扩展到了推理阶段。在推理过程中，我们通过对模型执行 M 次随机前向传播来近似后验预测分布，每次传播都使用从伯努利分布中独立采样的 dropout 掩码。此过程为每个输入实例生成一组电压预测结果。然后计算经验预测均值和方差，以近似预测分布并量化模型置信度。图 3 展示了这一过程，其中应用 M 个 dropout 掩码来输出相同输入 x 对应的目标 y 的预测分布。

IV. ES

我们对所提出的 PIGNN-3Φ模型在 18 节点改进型 CIGRE 低压不平衡配电馈线上进行了评估，具体细节见文献[14]。馈线布局如图 4 所示。为了生成训练数据集，我们采用了一种在功率因数计算（PF）研究中常用的蒙特卡罗（MC）采样策略。具体而言，首先从一组标准负荷曲线中采样出居民用户的有功功率和无功功率曲线。为了引入负荷模式中真实的波动性，我们对采样曲线施加了加性高斯噪声（标准差 15%），该噪声在各相和各时间步长上独立施加。随后，我们使用 pandapower 框架[15]对扰动后的负荷曲线进行数值功率因数仿真，以获得相应的电压标签。我们生成了 2400 个电网快照，并按 60/20/20 的比例将其划分为训练集、验证集和测试集。该模型使用 PyTorch 和 PyTorch Geometric 库[16]在 Python 中实现。为了进行基准测试，我们实现了两个黑盒神经网络模型：图神经网络（GNN）和全连接神经网络（FCNN）。这两个基线模型均包含：

5 隐藏层每层 128 个单元，采用 ReLU 激活函数。

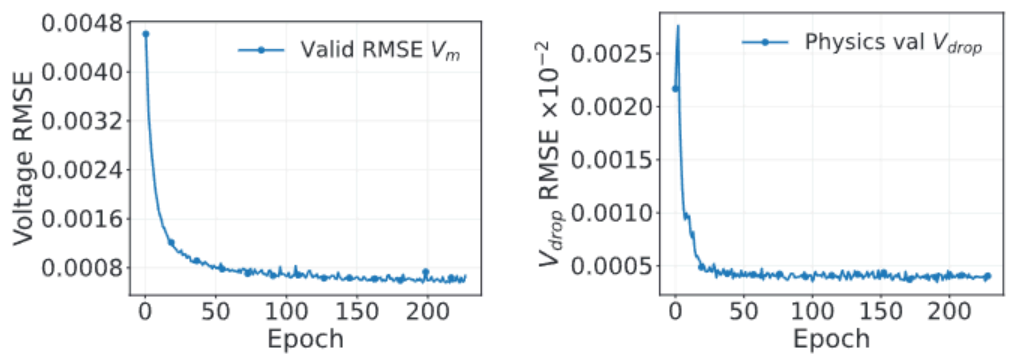
V. NR

本文提出的 PIGNN-3Φ模型与黑盒 GNN 和 FCNN 模型进行了比较，并以牛顿-拉夫逊 PF 解作为基准。比较评估了四个关键方面：（i）电压预测精度；（ii）MC dropout 不确定性估计的校准和质量；（iii）

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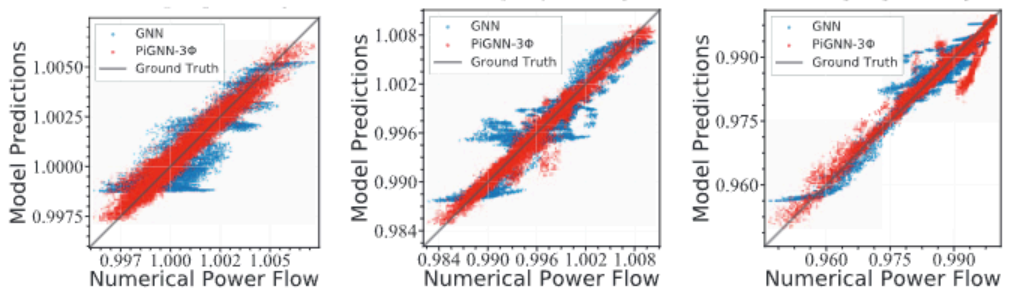
TABLE I: Performance comparison of different models. Lower RMSE and MAE, and higher R^2 indicate better performance.

Comparison Model	RMSE ($\times 10^{-3}$)				MAE ($\times 10^{-3}$)				R^2			
	Phase A	Phase B	Phase C	Overall	Phase A	Phase B	Phase C	Overall	Phase A	Phase B	Phase C	Overall
FCNN	1.100	2.200	2.600	2.000	0.900	1.600	2.000	1.500	0.780	0.840	0.930	0.850
GNN	0.830	1.910	2.310	1.680	0.660	1.410	1.710	1.260	0.805	0.856	0.942	0.867
PIGNN-3Φ	**0.450**	**0.930**	**2.170**	**1.180**	**0.340**	**0.620**	**1.160**	**0.710**	**0.945**	**0.966**	**0.949**	**0.953**



(a) Voltage RMSE (b) V_{drop} RMSE

Fig. 5: PIGNN-3Φ model validation set RMSE trends for voltages and V_{drop} (i.e., Ohm's law) during training.



(a) Phase A (b) Phase B (c) Phase C

Fig. 6: Comparing predicted voltages from both the GNN and PIGNN-3Φ models with ground truth across three phases.

robustness and generalization to previously unseen loading conditions; and (iv) inference time relative to iterative solvers. The model's accuracy is evaluated using several metrics, such as the coefficient of determination (R^2), the root mean square error (RMSE), and the mean absolute error (MAE). The R^2 metric quantifies how well the model's predictions align with true values by measuring the proportion of variance in the target explained by the model. The higher R^2 values indicate better model performance. Fig. 5 shows the decreasing validation RMSE trends over training epochs for both bus voltage magnitude and voltage drop (i.e., Ohm's law). These loss plots indicate that the PIGNN-3Φ model simultaneously learns data-fit and physics constraints, which results in stable training and fast convergence. In the following subsections, a detailed analysis of the experimental results is presented.

A. Voltage Prediction Accuracy

As shown in Table I, the PIGNN-3Φ model consistently achieves superior accuracy across all three evaluation metrics compared to the data-driven GNN and FCNN models. Fig. 6 presents scatter plots comparing the predicted and actual voltage magnitudes from the proposed PIGNN-3Φ and GNN for all three phases. The closer the data points are to the diagonal line in these plots, the better the predictive accuracy. The PIGNN-3Φ demonstrates better predictive performance

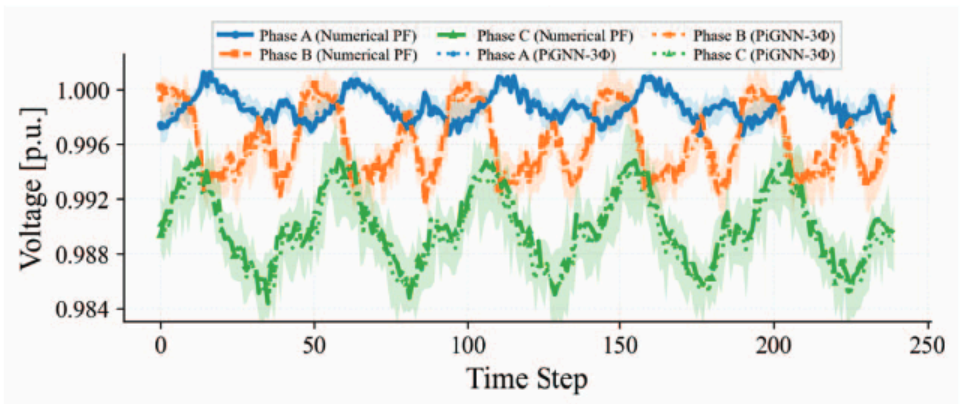
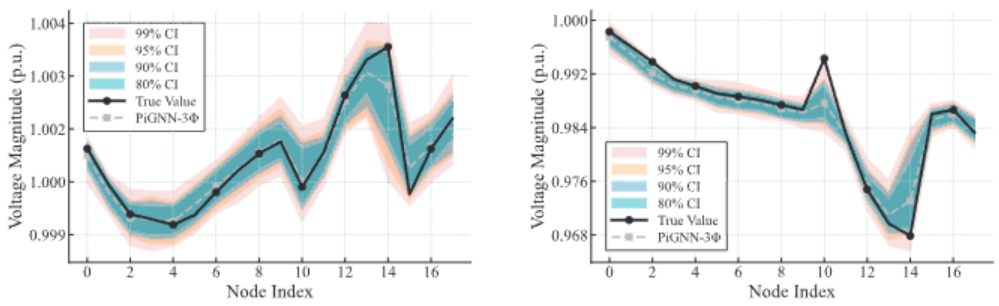


Fig. 7: Voltage predictions at Bus 3: Phase A (blue), Phase B (orange), and Phase C (green), with 95% confidence intervals (shaded regions). Solid lines indicate true values from the numerical solver; dashed lines represent PIGNN-3Φ predictions.



(a) Phase A (b) Phase C

Fig. 8: Calibrated predictive intervals generated by PIGNN-3Φ for voltage magnitudes across phases A and C of all nodes.

across three phases compared to the black-box GNN. This demonstrates the model's improved ability to capture complex phase-wise PF relationships and to learn phase unbalance more effectively. Moreover, the reduced dispersion in the predictions is largely due to the physics-informed constraints, which help the model reject physically implausible solutions and ensure consistency with PF principles. A sample probabilistic voltage prediction at bus 3 for each phase is shown in Fig. 7.

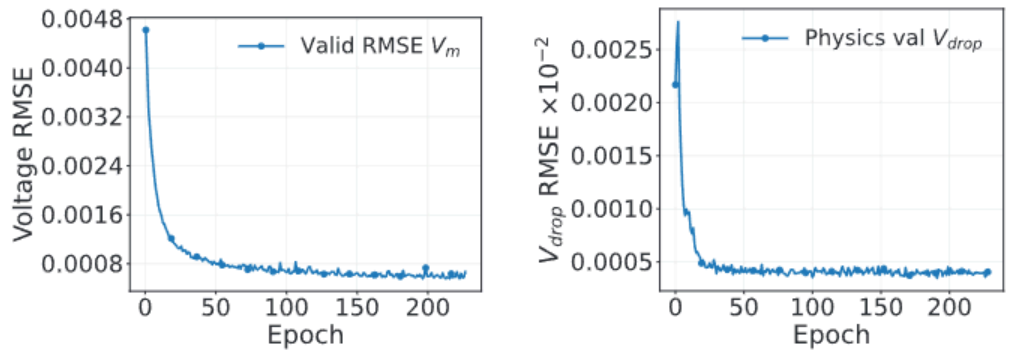
B. Calibrated Uncertainty under Chance Constraints

The calibration process in UQ is crucial because it ensures reliable interpretation of the model's predictive uncertainty. Calibration is performed on the validation set by comparing empirical coverage rates with nominal confidence levels. Using MC dropout, we compute multiple CIs for voltage predictions and calibrate them to align with the observed errors. Fig. 8 shows the calibrated intervals for all nodes. Narrower intervals (e.g., 80%) capture most true voltage values, indicating high predictive accuracy of the proposed model, while wider intervals (e.g., 99%) provide more conservative and robust coverage. Besides that, we also observed that the predictive intervals are systematically wider for nodes with very high loading or abrupt load variations (e.g., buses 12–14), indi-

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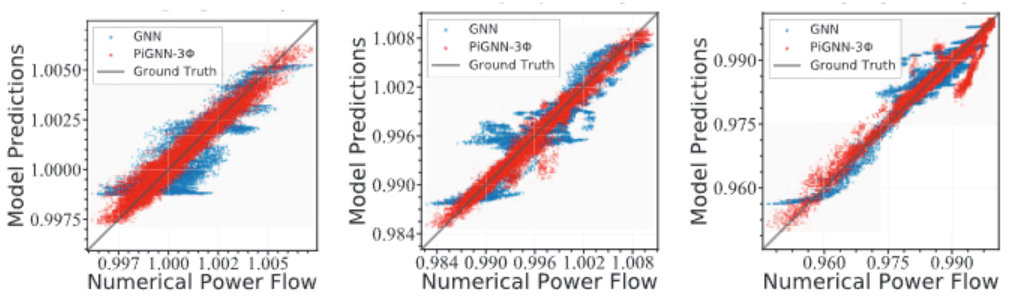
表一：不同模型的性能比较。RMSE 和 MAE 越低，Rin 值越高，表示性能越好。

比较	均方根误差（×10）				IS (×10)				R			
模型	A 阶段	B 阶段	C 阶段	全面的	A 阶段	B 阶段	C 阶段	全面的	A 阶段	B 阶段	C 阶段	全面的
全卷积神经网络	1.100	2.200	2.600	2.000	0.900	1.600	2.000	1.500	0.780	0.840	0.930	0.850
全球网络	0.830	1.910	2.310	1.680	0.660	1.410	1.710	1.260	0.805	0.856	0.942	0.867
PIGNN-3Φ	0.450	0.930	2.170	1.180	0.340	0.620	1.160	0.710	0.945	0.966	0.949	0.953



（a）电压均方根误差　　（b）VRMSE

图 5：PIGNN-3Φ 模型验证集 RMSE 趋势，电压和 V（即欧姆定律）在训练期间。



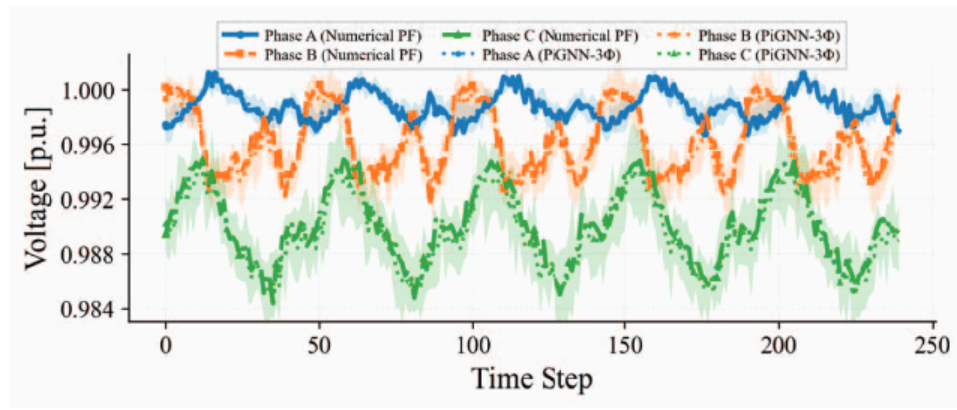
（a）A 阶段　　（b）B 阶段　　（c）C 阶段

图 6：比较 GNN 和 PIGNN-3Φ 模型预测的三相电压与真实值。

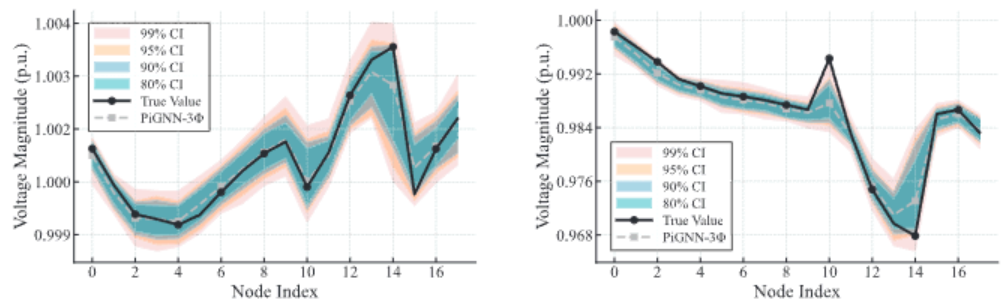
（iv）模型的鲁棒性和对先前未见过的负载条件的泛化能力；以及相对于迭代求解器的推理时间。模型的准确性使用多种指标进行评估，例如决定系数 (R)、均方根误差 (RMSE) 和平均绝对误差 (MAE)。R 指标通过测量模型解释的目标方差比例来量化模型预测与真实值的吻合程度。R 值越高，表明模型性能越好。图 5 显示了母线电压幅值和电压降（即欧姆定律）的验证 RMSE 随训练轮数的下降趋势。这些损失曲线表明，PIGNN-3Φ 模型能够同时学习数据拟合和物理约束，从而实现稳定的训练和快速收敛。以下小节将对实验结果进行详细分析。

A. 电压预测精度

如表 I 所示，与数据驱动的 GNN 和 FCNN 模型相比，PIGNN-3Φ 模型在所有三个评估指标上均表现出更优的精度。图 6 展示了所提出的 PIGNN-3Φ 和 GNN 模型对所有三相电压幅值的预测值与实际值的散点图。图中数据点越接近对角线，预测精度越高。



如图 7 所示，PIGNN-3Φ 展现出更优的预测性能：母线 3 的电压预测结果：A 相（蓝色）、B 相（橙色）和 C 相（绿色），以及 95% 置信区间（阴影区域）。实线表示数值求解器得到的真实值；虚线表示 PIGNN-3Φ 的预测值。



（a）A 阶段　　（b）C 阶段

图 8：PIGNN-3Φ 为所有节点 A 相和 C 相电压幅值生成的校准预测区间。

与黑箱图神经网络（GNN）相比，该模型在三个相位上均表现出更优异的性能。这表明该模型能够更好地捕捉复杂的相位功率因数关系，并更有效地学习相位不平衡。此外，预测结果离散度的降低主要归功于物理约束，这些约束有助于模型排除物理上不合理的解，并确保与功率因数原理的一致性。图 7 展示了母线 3 各相位的概率电压预测示例。

B. 概率约束下的校准不确定性

在不确定性量化（UQ）中，校准过程至关重要，因为它确保了对模型预测不确定性的可靠解释。校准是在验证集上进行的，方法是将经验覆盖率与标称置信水平进行比较。利用蒙特卡罗（MC）dropout，我们计算了电压预测的多个置信区间（CI），并对其进行校准以使其与观测误差保持一致。图 8 显示了所有节点的校准区间。较窄的区间（例如 80%）能够捕捉到大部分真实电压值，表明所提出的模型具有较高的预测精度；而较宽的区间（例如 99%）则提供了更保守、更稳健的覆盖范围。此外，我们还观察到，对于负载非常高或负载突变较大的节点（例如母线 12-14），预测区间系统性地更宽。

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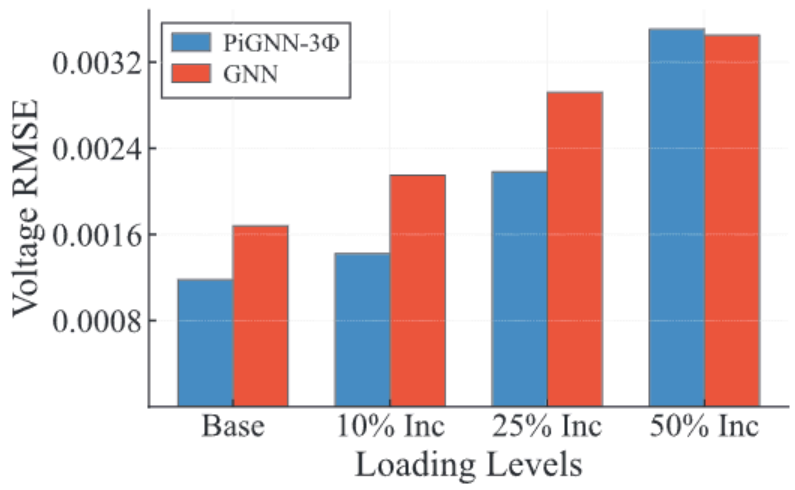


Fig. 9: Comparison between proposed PIGNN-3Φ and GNN model under different network loading conditions.

cating higher model uncertainty. By selecting an appropriate confidence level, operators can enforce probabilistic voltage limits that bound the probability of violations to a prescribed risk tolerance, thereby supporting risk-aware decision making.

C. Impact of Loading Variations

To assess the model's robustness under varying network loading conditions, we increased the active and reactive power injections at each bus by 10%–50% above their nominal values. The model, trained only on nominal load levels, was evaluated on unseen high-loading scenarios to further assess its generalization ability. As illustrated in Fig. 9, the RMSE of the proposed PIGNN-3Φ model increases less than that of the black-box GNN as loading levels increase. For minor load changes (+10%), the PIGNN-3Φ maintains high accuracy with minimal RMSE increase. However, for significant load changes (+25% and +50%), the RMSE rises more noticeably, reflecting the challenges of generalizing to extreme conditions. The proposed model handles moderate load variations well, but is more sensitive to heavy load scenarios; therefore, retraining might be needed for very high loading conditions.

D. Time Complexity

The PIGNN-3Φ model offers a significantly lower computational cost than traditional numerical PF solvers, making it well-suited for performing large-scale network expansion and planning studies. We compared the proposed model inference time with the numerical solver on two differently sized networks: the CIGRE 18-bus and the IEEE LV European 906-bus network. The numerical solver scaled poorly with increasing network size (i.e., 56.7 ms per sample for the 18-bus and 561 ms for the 906-bus). In contrast, PIGNN-3Φ achieves much better computational efficiency, with per sample inference time increased only marginally with network size (i.e., 4.5 ms for the 18-bus and 11.4 ms for the 906-bus), which provides up to a 50× speed-up. This experiment was performed on a machine with an Intel Core i7 CPU (16 cores) and 32 GB of RAM.

VI. Conclusion

We propose a physics-informed graph neural network for performing fast probabilistic power flow calculations in three-phase unbalanced distribution networks. By embedding constraints based on DistFlow equations into the objective function, the model ensures physically consistent predictions and effectively captures phase unbalance. We incorporated the MC dropout approach to enable probabilistic inference with uncertainty quantification, which provides reliable confidence intervals for risk-aware decision making. Experimental results show that the proposed model achieves accuracy comparable to traditional Newton-Raphson. Future work will focus on evaluating this approach on larger networks under partial observability, along with topological variations.

References

[1] B. Sereeter, K. Vuik, and C. Witteveen, "Newton power flow methods for unbalanced three-phase distribution networks," *Energies*, vol. 10, no. 10, 2017.

[2] V. Bassi, L. F. Ochoa, T. Alpcan, and C. Leckie, "Electrical model-free voltage calculations using neural networks and smart meter data," *IEEE Transactions on Smart Grid*, 2022.

[3] A. S. Zamzam and K. Baker, "Learning optimal solutions for extremely fast ac optimal power flow," in *2020 IEEE International Conference on Communications, Control, and Computing Technologies for Smart Grids (SmartGridComm)*, pp. 1–6, IEEE, 2020.

[4] J. B. Hansen, S. N. Anfinsen, and F. M. Bianchi, "Power flow balancing with decentralized graph neural networks," *IEEE Transactions on Power Systems*, 2022.

[5] A. B. Jeddi and A. Shafieezadeh, "A physics-informed graph attention-based approach for power flow analysis," in *2021 20th IEEE International conference on machine learning and applications (ICMLA)*, pp. 1634–1640, IEEE, 2021.

[6] F. Azam, C. Hermans, T. Becker, R. D'hulst, K. Vanthournout, and G. Deconinck, "Accelerating power flow calculations in LV networks using physics-informed graph neural networks," in *ISGT Conference, Belgrade 2024*, pp. 1–5, 2024.

[7] L. Pagnier and M. Chertkov, "Physics-informed graphical neural network for parameter & state estimations in power systems," *arXiv preprint arXiv:2102.06349*, 2021.

[8] J. Ostrometzky, K. Berestizshevsky, A. Bernstein, and G. Zussman, "Physics-informed deep neural network method for limited observability state estimation," *Electrical Engineering and Systems Science*, 12 2019.

[9] X. Hu, H. Hu, S. Verma, and Z.-L. Zhang, "Physics-guided deep neural networks for power flow analysis," *IEEE Transactions on Power Systems*, vol. 36, pp. 2082–2092, 2020.

[10] N. Lin, S. Orfanoudakis, N. O. Cardenas, J. S. Giraldo, and P. P. Vergara, "Powerflownet: power flow approximation using message passing graph neural networks," *International Journal of Electrical Power Energy Systems*, vol. 160, p. 110112, 2024.

[11] Y. Gal and Z. Ghahramani, "Dropout as a Bayesian approximation: representing model uncertainty in deep learning," in *Proceedings of The 33rd International Conference on Machine Learning*, vol. 48 of *Proceedings of Machine Learning Research*, (NY, USA), pp. 1050–1059, 2016.

[12] T. Su, J. Zhao, Y. Pei, and F. Ding, "Probabilistic physics-informed graph convolutional network for active distribution system voltage prediction," *IEEE Transactions on Power Systems*, vol. 38, no. 6, pp. 5969–5972, 2023.

[13] M. Baran and F. Wu, "Network reconfiguration in distribution systems for loss reduction and load balancing," *IEEE Transactions on Power Delivery*, vol. 4, no. 2, pp. 1401–1407, 1989.

[14] T. Antić, L. Thurner, T. Capuder, and I. Pavić, "Modeling and open source implementation of balanced and unbalanced harmonic analysis in radial distribution networks," *Electric Power Systems Research*, vol. 209, p. 107935, 2022.

[15] L. Thurner, A. Scheidler, F. Schäfer, J.-H. Menke, J. Dollichon, F. Meier, S. Meinecke, and M. Braun, "Pandapower—an open source Python tool for convenient modeling, analysis, and optimization of electric power systems," *IEEE Transactions on Power Systems*, vol. 33, pp. 6510–6521, 2017.

[16] M. Fey and J. E. Lenssen, "Fast graph representation learning with PyTorch Geometric," *ArXiv*, vol. abs/1903.02428, 2019.

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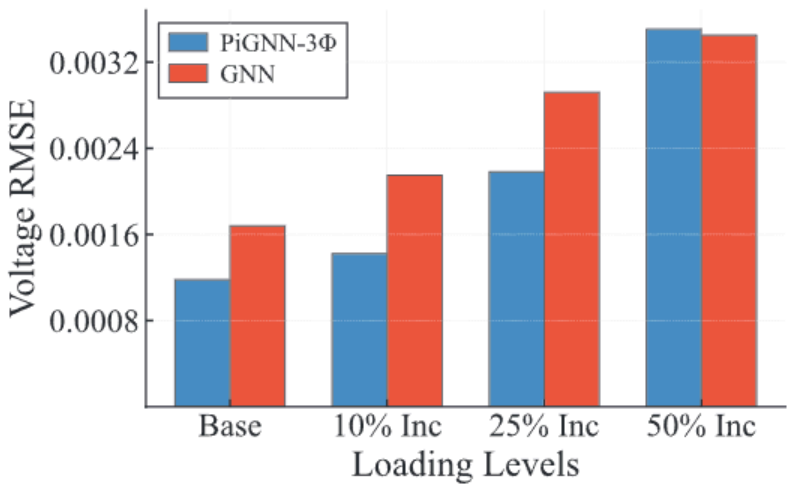


图 9：在不同网络负载条件下，所提出的 PIGNN-3Φ 模型与 GNN 模型的比较。

考虑到模型的不确定性较高，运营商可以通过选择合适的置信水平来实施概率电压限制，将违规概率限制在预设的风险承受范围内，从而支持风险感知决策。

C. 载荷变化的影响

为了评估模型在不同网络负载条件下的鲁棒性，我们将每个母线的有功功率和无功功率注入量分别增加到标称值的 10%至 50%。该模型仅在标称负载水平下进行训练，并在未见过的高负载场景下进行评估，以进一步评估其泛化能力。如图 9 所示，随着负载水平的增加，所提出的 PIGNN-3Φ模型的均方根误差（RMSE）增幅小于黑盒 GNN 模型。对于较小的负载变化（+10%），PIGNN-3Φ模型保持了较高的精度，RMSE 增幅极小。然而，对于较大的负载变化（+25%和+50%），RMSE 的增幅更为明显，这反映了模型在极端负载条件下泛化的挑战。所提出的模型能够很好地处理中等负载变化，但对重负载场景更为敏感；因此，在极高的负载条件下可能需要重新训练模型。

D. 时间复杂度

与传统的数值 PF 求解器相比，PIGNN-3Φ模型具有显著更低的计算成本，使其非常适合用于大规模网络扩展和规划研究。我们在两个不同规模的网络上比较了所提出的模型与数值求解器的推理时间：CIGRE 18 节点网络和 IEEE LV European 906 节点网络。数值求解器的性能随着网络规模的增加而显著下降（例如，18 节点网络每个样本的推理时间为 56.7 毫秒，906 节点网络为 561 毫秒）。相比之下，PIGNN-3Φ模型实现了更高的计算效率，每个样本的推理时间仅随网络规模的增加而略有增加（例如，18 节点网络为 4.5 毫秒，906 节点网络为 11.4 毫秒），速度提升高达 50 倍。该实验在一台配备 Intel Core i7 CPU（16 核）和 32 GB 内存的机器上进行。

六、C

我们提出了一种基于物理信息的图神经网络，用于在三相不平衡配电网络中快速执行概率潮流计算。通过将基于 DistFlow 方程的约束嵌入目标函数，该模型确保了物理上一致的预测结果，并有效捕捉了相位不平衡现象。我们引入了 MC dropout 方法，实现了不确定性量化的概率推理，从而为风险感知决策提供了可靠的置信区间。实验结果表明，所提出的模型达到了与传统牛顿-拉夫逊算法相当的精度。未来的工作将着重于在部分可观测性以及拓扑结构变化的情况下，评估该方法在大规模网络上的性能。

R

[1] B. Sereeter、K. Vuik 和 C. Witteveen，“不平衡三相配电网络的牛顿潮流计算方法”，《能源》（Energies），第 10 卷，第 10 期，2017 年。[2] V. Bassi、LF Ochoa、T. Alpcan 和 C. Leckie，“基于神经网络和智能电表数据的无模型电压计算”，《IEEE 智能电网汇刊》（IEEE Transactions on Smart Grid），2022 年。[3] AS Zamzam 和 K. Baker，“学习极高电压问题的最优解”。

快速交流最优潮流”，载于 2020 年 IEEE 智能电网通信、控制和计算技术国际会议 (SmartGridComm)，第 1-6 页，IEEE，2020 年。

[4] JB Hansen、SN Anfinsen 和 FM Bianchi，“利用分散式图神经网络实现潮流平衡”，IEEE 电力系统汇刊，2022 年。

[5] AB Jeddi 和 A. Shafieezadeh，“一种基于物理信息的图注意力法的潮流分析方法”，发表于 2021 年第 20 届 IEEE 国际电力系统工程会议论文集。

全国机器学习与应用会议（ICMLA），第 1634-1640 页，IEEE，2021 年。

[6] F. Azam、C. Hermans、T. Becker、R. D'hulst、K. Vanthournout 和 G. Deconinck，“使用物理信息图神经网络加速低压网络中的潮流计算”，ISGT 会议，贝尔格莱德 2024，第 1-5 页，2024 年。[7] L. Pagnier 和 M. Chertkov，“用于电力系统参数和状态估计的物理信息图神经网络”，arXiv 预印本 arXiv:2102.06349，2021 年。

[8] J. Ostrometzky、K. Berestizshevsky、A. Bernstein 和 G. Zussman，“基于物理信息的深度神经网络方法用于有限可观测性状态估计”，电气工程与系统科学，2019 年 12 月。[9] X. Hu、H. Hu、S. Verma 和 Z.-L. Zhang，“用于潮流分析的物理引导深度神经网络”，IEEE 电力系统汇刊，第 36 卷，第 2082-2092 页，2020 年。

[10] N. Lin、S. Orfanoudakis、NO Cardenas、JS Giraldo 和 PP Vergara，“Powerflownet：基于消息传递图的潮流近似
神经网络”，《国际电力能源系统杂志》，第 160 卷，第 110112 页，2024 年。

[11] Y. Gal 和 Z. Ghahramani，“Dropout 作为贝叶斯近似：
“在深度学习中表示模型不确定性”，载于《第 33 届国际机器学习会议论文集》，机器学习研究论文集第 48 卷（美国纽约），第 1050-1059 页，2016 年。

[12] T. Su、J. Zhao、Y. Pei 和 F. Ding，“用于主动配电系统电压预测的概率物理信息图卷积网络”，
IEEE 电力系统学报，第 38 卷，第 6 期，第 5969-5972 页，2023 年。

[13] M. Baran 和 F. Wu，“配电系统中的网络重构”
为了减少损耗和平衡负载”，IEEE 电力输送学报，第 4 卷，第 2 期，第 1401-1407 页，1989 年。

[14] T. Anti´c、L. Thurner、T. Capuder 和 I. Pavi´c，“建模和开放
“辐射状配电网络中平衡和不平衡谐波分析的源实现”，《电力系统研究》，第 209 卷，第 107935 页，2022 年。

[15] L.瑟纳，A.谢德勒，F.谢弗，J.-H。 Menke、J. Dollichon、F. Meier、S. Meinecke 和 M. Braun，“Pandapower——一个用于方便地对电力系统进行建模、分析和优化的开源 Python 工具”，IEEE 电力系统学报，第 33 卷，第 6510-6521 页，2017 年。

[16] M. Fey 和 JE Lenssen，“使用 PyTorch Geometric 进行快速图表示学习”，ArXiv，卷 abs/1903.02428，2019 年。

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